

KAPREKAR CONTEST – FINAL – SUB-JUNIOR

(Classes VII and VIII)

AMTI – 31ST OCTOBER, 2015.



- Instructions:
1. Answer as many questions as possible.
 2. Elegant and novel solutions will get extra credits.
 3. Diagrams and explanations should be given wherever necessary.
 4. Fill in FACE SLIP and your rough working should be in the answer book itself.
 5. Maximum time allowed is THREE hours.
 6. All questions carry equal marks.

1. a)

	175										70	
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The diagram above contains 13 boxes. The numbers in the second and the twelfth boxes are respectively 175 and 70. Fill up the boxes with natural numbers such that

- (i) sum of all numbers in all the 13 boxes is 2015.
- (ii) sum of the numbers in any three consecutive boxes is always the same.

The solution must contain the steps how you arrive at the numbers.

b) if x, y, z are real and unequal numbers, prove that

$$2015x^2 + 2015y^2 + 6z^2 > 2(2012xy + 3yz + 3zx)$$

2. If a, b, c are reals such that $a + b = 4$ and $2c^2 - ab = 4\sqrt{3}c - 10$, find the numerical values of a, b and c .

3. When $a = 2^{2014}$ and $b = 2^{2015}$, prove that

$$\left\{ \frac{\frac{(a+b)^2 + (a-b)^2}{b-a} - (a+b)}{\frac{1}{b-a} - \frac{1}{a+b}} \right\} \div \left\{ \frac{(a+b)^3 + (b-a)^3}{(a+b)^2 - (a-b)^2} \right\} \text{ is divisible by 3.}$$

4. Prove that the feet of the perpendiculars drawn from the vertices of a parallelogram on to its diagonals are the vertices of a parallelogram.

5. ABC is an acute angled triangle. P, Q are the points on AB and AC respectively such that area of $\Delta APC =$ area of ΔAQB . A line is drawn through B parallel to AC and meets the line through Q parallel to AB at S . QS cuts BC at R . Prove that $RS = AP$.

6. a) A man is walking from a town A to another town B at a speed of 4km/hr. He started one hour before a bus starts. The bus is travelling with a speed of 12km/hr. The man on the way got into the bus and travels 2 hours and reached town B . What is the distance between town A and town B .

b) A point P is taken within a rhombus $ABCD$ such that $PA = PC$. Show that B, P, D are collinear.

7. If $(x + y + z)^3 = (y + z - x)^3 + (z + x - y)^3 + (x + y - z)^3 + kxyz$ find the numerical value of k . Deduce the following result.

If $a = 2015, b = 2014, c = \frac{1}{2014}$ prove that

$$(a + b + c)^3 - (a + b - c)^3 - (b + c - a)^3 - (c + a - b)^3 - 23abc = 2015$$